Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Duarte Acres Holsteins

Physical address of dairy:

8413 Faith Home RD	Ceres	Stanislaus	95307
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1958

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X022-X018-X005-X000

B. OPERATORS

Nelson, Dwayne

Operator name: Nelson, Dwayne

Telephone no.: (209) 678-1151 (Landline) (Cellular)

5931 W Tuolumne RD	Ceres	CA	95307
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Duarte, Eileen

Legal owner name: Duarte, Eileen

Telephone no.: (209) 632-1987 (Landline) (Cellular)

8413 Faith Home RD	Ceres	CA	95307
Mailing Address Number and Street	City	State	Zip Code

Nelson, Irene

Legal owner name: Nelson, Irene

Telephone no.: (209) 613-3288 (Landline) (Cellular)

5931 W Tuolumne RD	Ceres	CA	95307
Mailing Address Number and Street	City	State	Zip Code

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	0	0	65	60	0	0
Maximum number	0	0	65	65	0	0
Average number	0	0	65	60	0	0
Avg live weight (lbs)	0	0	1,100	900		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 1,308.34 *tons per reporting period*

Total nitrogen from manure: 11,862.50 *lbs per reporting period*

After ammonia losses (30% loss applied): 8,303.75 *lbs per reporting period*

Total phosphorus from manure: 2,007.50 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 9,300,000 *gallons*

Total nitrogen generated: 6,757.45 *lbs*

Total phosphorus generated: 1,270.28 *lbs*

Total potassium generated: 18,959.01 *lbs*

Total salt generated: 138,443.55 *lbs*

	9,300,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	9,300,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Deep Well	Ground water
Domestic Supply Well	Ground water
TID Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

Date	Material type / Description	Quantity	Reporting basis	Moist. (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	Corral solids *Corral Solids*	459.00 *ton*	As-is	14.0	17,100.00	4,300.00	6,260.00		0.00

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	15,697.80	3,947.40	5,746.68	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	15,697.80	3,947.40	5,746.68	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field #4	40	40	2	both	X022-X018-X004-X000
Totals for areas that were used for application	40	40	2		
Totals for areas that were not used for application					
Land application area totals	40	40	2		

B. CROPS AND HARVESTS

Field #4

Field name: Field #4

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	327.50 *ton*	Dry-weight		69.2	24,900.00	5,090.00	47,800.00		12.18

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	8.19	125.58	25.67	241.08	614.30

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/02/2023	1,274.39 *ton*	Dry-weight		64.4	8,900.00	1,390.00	14,900.00		5.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.86	201.89	31.53	337.99	1,202.26

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field #4 - 11/02/2022: Oats, silage-soft dough

Field name: Field #4

Crop: Oats, silage-soft dough — Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Plow/disc	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	44.10	16.40	46.90	0.00	100.00 *ton*
Application event totals		44.10	16.40	46.90	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/16/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	11.99	4.03	41.23	180.25	1,080,000.00 *gal*
Deep Well	Ground water	65.38	0.00	0.00	1,083.09	7,271,064.91 *gal*
Application event totals		77.37	4.03	41.23	1,263.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/02/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	7.99	2.69	27.49	120.17	720,000.00 *gal*
Deep Well	Ground water	43.59	0.00	0.00	722.06	4,847,376.60 *gal*
Application event totals		51.58	2.69	27.49	842.22	

Field #4 - 06/10/2023: Corn, silage

Field name: Field #4

Crop: Corn, silage — Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field #4 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	20.85	3.50	56.74	442.49	1,050,000.00 *gal*
TID Canal	Surface water	0.38	0.00	0.00	35.39	7,069,090.88 *gal*
Application event totals		21.24	3.50	56.74	477.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Deep Well	Ground water	47.22	0.00	0.00	782.23	5,251,324.65 *gal*
TID Canal	Surface water	0.02	0.00	0.00	2.02	403,948.05 *gal*
Application event totals		47.24	0.00	0.00	784.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	20.85	3.50	56.74	442.49	1,050,000.00 *gal*
TID Canal	Surface water	0.22	0.00	0.00	20.23	4,039,480.50 *gal*
Application event totals		21.07	3.50	56.74	462.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	20.85	3.50	56.74	442.49	1,050,000.00 *gal*
TID Canal	Surface water	0.22	0.00	0.00	20.23	4,039,480.50 *gal*
Application event totals		21.07	3.50	56.74	462.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	20.85	3.50	56.74	442.49	1,050,000.00 *gal*
TID Canal	Surface water	0.22	0.00	0.00	20.23	4,039,480.50 *gal*
Application event totals		21.07	3.50	56.74	462.72	

Field #4 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	20.85	3.50	56.74	442.49	1,050,000.00 *gal*
TID Canal	Surface water	0.20	0.00	0.00	18.20	3,635,532.45 *gal*
Application event totals		21.05	3.50	56.74	460.70	

08/29/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	20.85	3.50	56.74	442.49	1,050,000.00 *gal*
TID Canal	Surface water	0.19	0.00	0.00	17.19	3,433,558.43 *gal*
Application event totals		21.04	3.50	56.74	459.69	

09/08/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	11.92	2.00	32.42	252.85	600,000.00 *gal*
TID Canal	Surface water	0.21	0.00	0.00	19.21	3,837,506.48 *gal*
Application event totals		12.12	2.00	32.42	272.07	

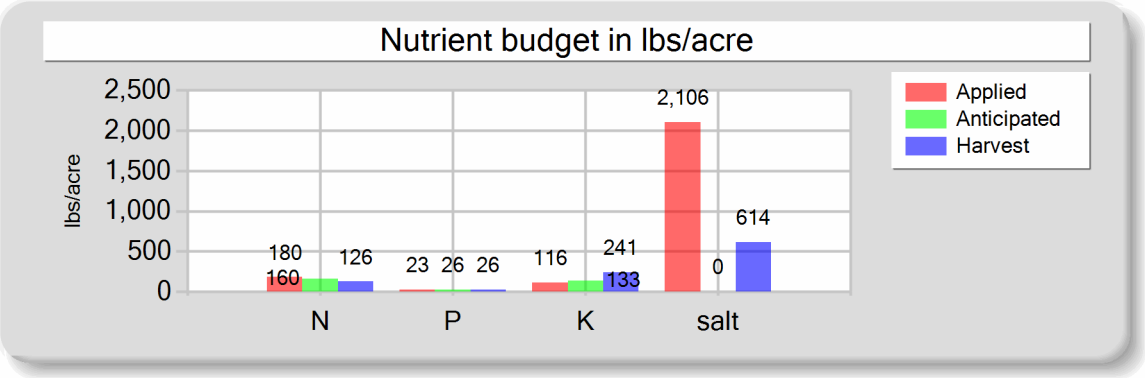
09/19/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	11.92	2.00	32.42	252.85	600,000.00 *gal*
TID Canal	Surface water	0.13	0.00	0.00	12.14	2,423,688.30 *gal*
Application event totals		12.05	2.00	32.42	264.99	

B. NUTRIENT BUDGET

Field #4 - 11/02/2022: Oats, silage-soft dough

Field name: Field #4 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	44.10	16.40	46.90	0.00
Process wastewater	19.98	6.72	68.72	300.42
Fresh water	108.97	0.00	0.00	1,805.14
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	180.04	23.12	115.62	2,105.56
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	125.58	25.67	241.08	614.30
Nutrient balance	54.46	-2.55	-125.46	1,491.26
Applied to removed ratio	1.43	0.90	0.48	3.43

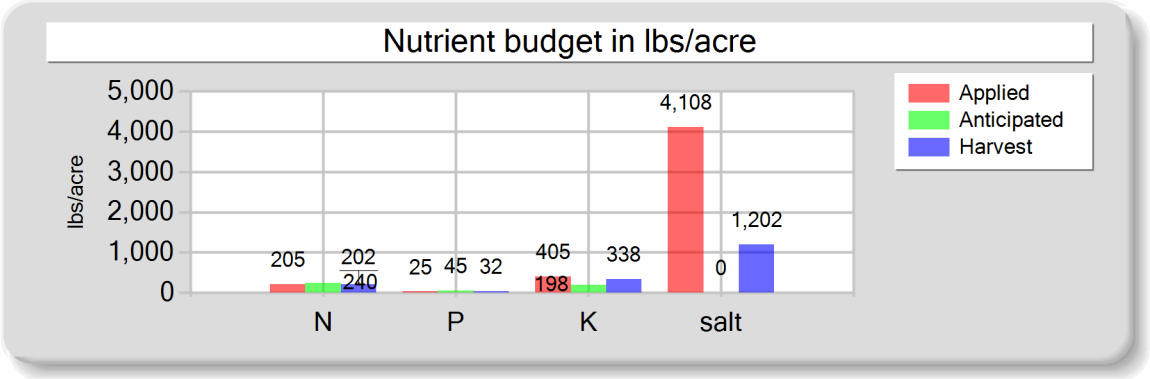
Fresh water applied	
12,118,441.51	*gallons*
446.28	*acre-inches*
11.16	*inches/acre*

Process wastewater applied	
1,800,000.00	*gallons*
66.29	*acre-inches*
1.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #4 - 06/10/2023: Corn, silage

Field name: Field #4 Crop: Corn, silage Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	148.96	25.04	405.25	3,160.67
Fresh water	49.00	0.00	0.00	947.07
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	204.96	25.04	405.25	4,107.74
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	201.89	31.53	337.99	1,202.26
Nutrient balance	3.07	-6.50	67.26	2,905.48
Applied to removed ratio	1.02	0.79	1.20	3.42

Fresh water applied	
38,173,090.74	*gallons*
1,405.79	*acre-inches*
35.14	*inches/acre*

Process wastewater applied	
7,500,000.00	*gallons*
276.20	*acre-inches*
6.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Manure Corral Solids

Sample and source description: Manure Corral Solids
Sample date: 01/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 33.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,800.00	6,830.00	31,100.00							
DL	0.01	0.01	0.01							

Manure Corral Solids

Sample and source description: Manure Corral Solids
Sample date: 04/10/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 14.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,100.00	4,300.00	6,260.00							
DL	0.01	0.01	0.01							

Manure

Sample and source description: Manure
Sample date: 10/03/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 6.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,700.00	4,560.00	15,700.00							
DL	0.01	0.01	0.01							

Manure

Sample and source description: Manure

Sample date: 12/31/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 21.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,820.00	3,280.00	9,380.00							
DL	0.01	0.01	0.01							

B. PROCESS WASTEWATER ANALYSES

Lagoon

Sample and source description: Lagoon

Sample date: 04/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	95.20	47.60			16.00	259.00								2,180.00	2,020
DL	0.01	0.01			0.01	0.01								0.01	1

Lagoon

Sample and source description: Lagoon

Sample date: 06/06/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	33.60	19.60			7.76	106.00								1,080.00	1,940
DL	0.01	0.01			0.01	0.01								0.01	1

Lagoon

Sample and source description: Lagoon

Sample date: 10/03/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	28.00	19.60			8.90	81.20								699.00	200
DL	0.01	0.01			0.01	0.01								0.01	1

Lagoon

Sample and source description: Lagoon

Sample date: 12/31/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	53.20	16.80			17.90	183.00								1,350.00	800
DL	0.01	0.01			0.01	0.01								0.01	1

C. FRESH WATER ANALYSES

Deep Well

Ag Well

Sample description: Ag Well

Sample date: 09/06/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.50	42.60								1,190.00	
DL		0.01	0.01								0.01	

Domestic Supply Well

Domestic Supply Well

Joe Duarte #1

Sample description: Joe Duarte #1

Sample date: 09/06/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.50	31.50								1,010.00	
DL		0.01	0.01								0.01	

Joe Duarte #2

Sample description: Joe Duarte #2

Sample date: 09/06/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.50	21.70								729.00	
DL		0.01	0.01								0.01	

TID Canal

Canal

Sample description: Canal

Sample date: 06/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.26		0.26								40.00	
DL	0.01		0.01								0.01	

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field #4 - 11/02/2022: Oats, silage-soft dough

Field #4 Oats

Sample and source description: Field #4 Oats

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,900.00	5,090.00	47,800.00		12.18
DL	0.01	0.01	0.01		0.01

Field #4 - 06/10/2023: Corn, silage

Field #4 Corn

Sample and source description: Field #4 Corn

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,900.00	1,390.00	14,900.00		5.30
DL	0.01	0.01	0.01		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

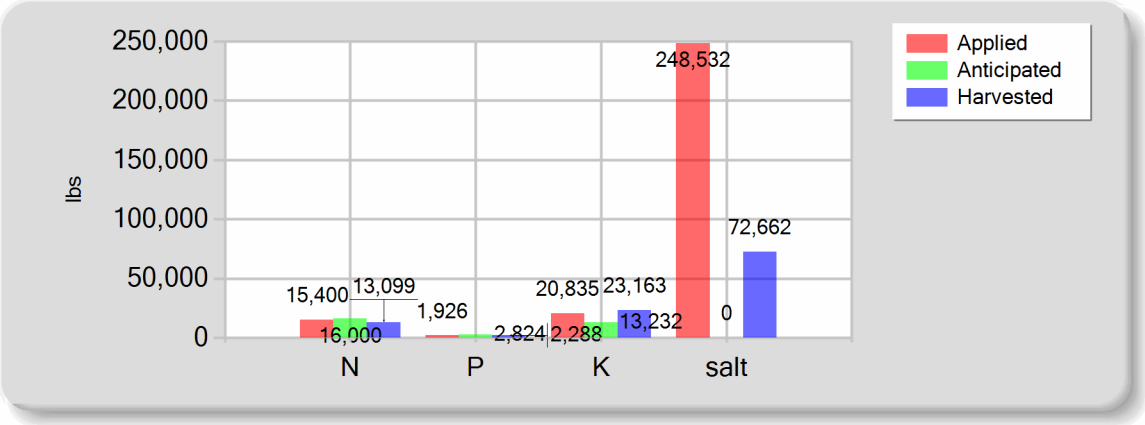
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

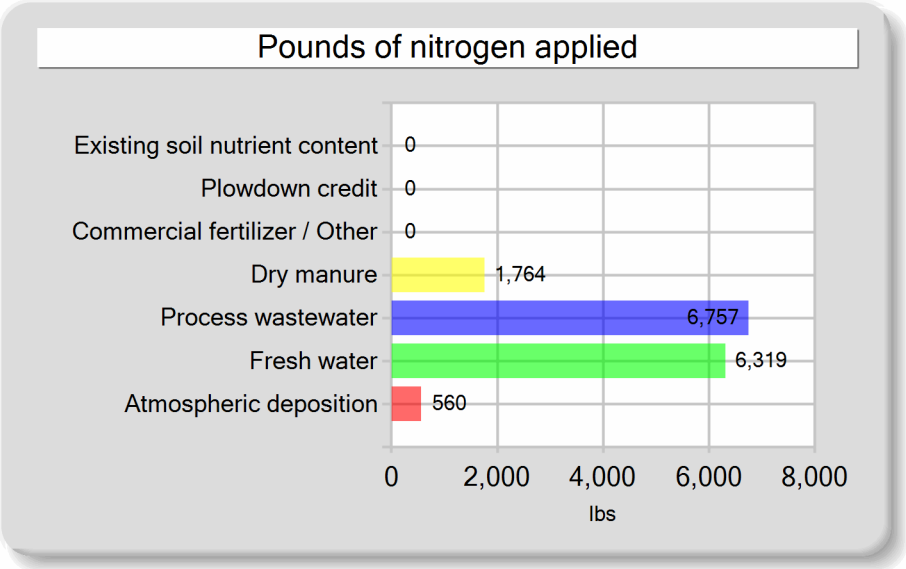
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

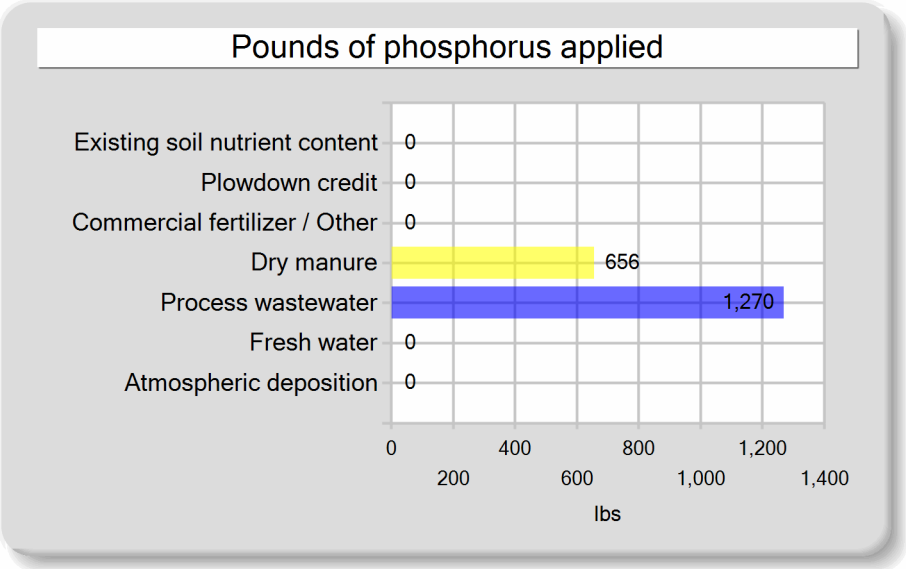
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	1,764.00	656.00	1,876.00	0.00
Process wastewater	6,757.45	1,270.28	18,959.01	138,443.55
Fresh water	6,318.81	0.00	0.00	110,088.37
Atmospheric deposition	560.00	0.00	0.00	0.00
Total nutrients applied	15,400.25	1,926.28	20,835.01	248,531.92
Anticipated crop nutrient removal	16,000.00	2,824.00	13,232.00	0.00
Actual crop nutrient removal	13,098.88	2,288.09	23,162.92	72,662.31
Nutrient balance	2,301.37	-361.82	-2,327.92	175,869.61
Applied to removed ratio	1.18	0.84	0.90	3.42

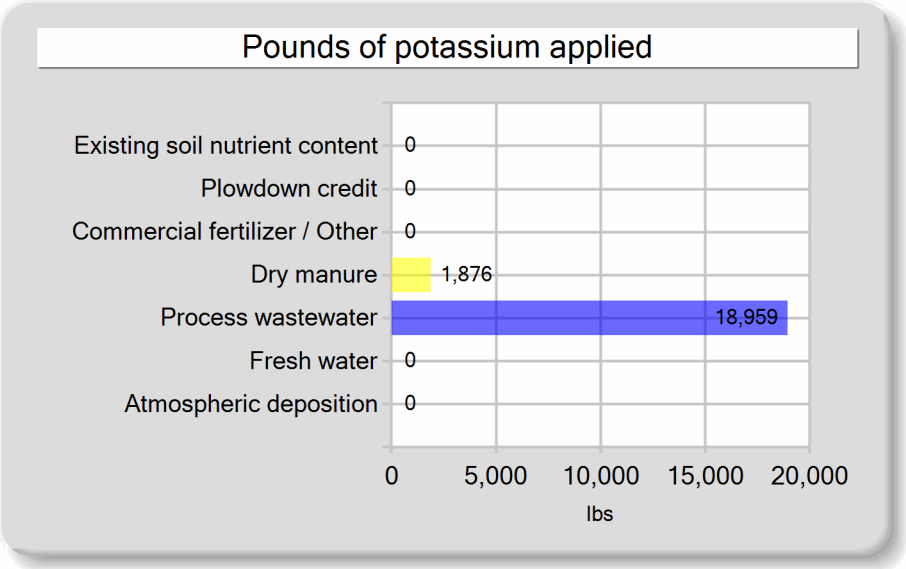
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

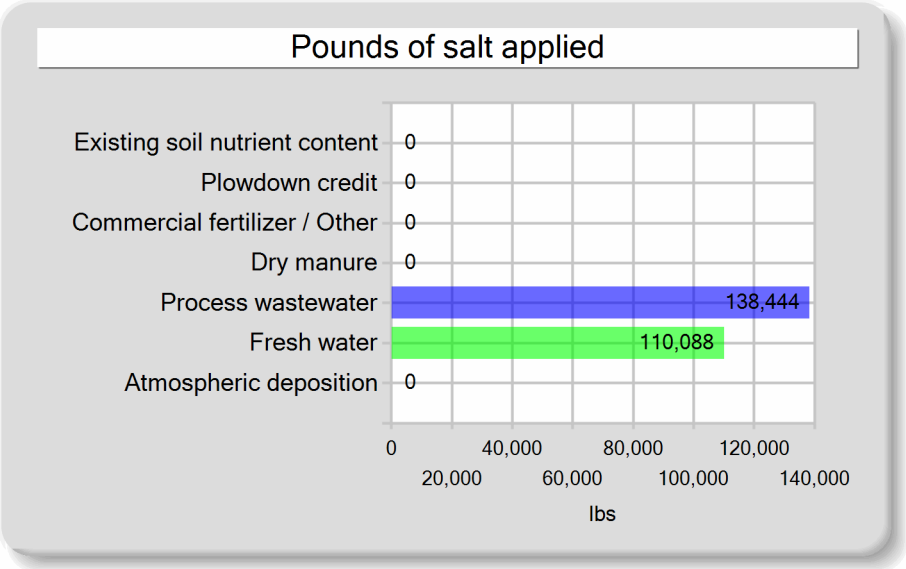


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Irene Nelson	Dwayne Nelson
PRINT OR TYPE NAME	PRINT OR TYPE NAME
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Irene Nelson	Dwayne Nelson
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-30-2024	6-30-2024
DATE	DATE

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Authorized Copy Released To:

Client: Nelson Farms
Mailing address: 5931 W Tuolumne Rd
(not facility address) Ceres CA 95307
Phone#: 209 678-1151

Sampled By: Duwayne Nelson
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	9-5-23	1 pm	Joe Duarte #1	domestic	11.8°		X				
2			Joe Duarte #2		11.8°						
3			Nelson Farms #1		11.6°						
4			Nelson Farms #2		11.6°						
5			Deep Well	Ag well	11.8°						
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	Duwayne Nelson	Nelson Farms	12:17 9-6-23			T3249019
2nd		DENELE ANALYTICAL, INC.	9-6-23 12:20			

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Nelson Farms
5931 W Tuolumne Rd
Ceres, CA 95307

Grower

Consultant:
Central Valley Ag Consulting
P.O. Box 582085
Modesto, CA 95358

PURCHASE ORDER:
RECEIVED DATE: 9/6/23 12:20 PM
SUBMITTED BY: Irene
ANALYZED DATE: 9/7/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Joe Duarte #1	T3249019A	Electrical Conductivity (EC)	EPA 120.1 and	1.01	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	31.5	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	643	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.6	95.8 (T3249019A)	93.4 (T3249019A)	2.53
Electrical Conductivity (EC)	94.4			0.100
Nitrate Nitrogen (NO3-N)	100	95.6 (W3248007A)	92.9 (W3248007A)	2.88

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results. nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Nelson Farms
5931 W Tuolumne Rd
Ceres, CA 95307

Grower

Consultant:
Central Valley Ag Consulting
P.O. Box 582085
Modesto, CA 95358

PURCHASE ORDER:
RECEIVED DATE: 9/6/23 12:20 PM
SUBMITTED BY: Irene
ANALYZED DATE: 9/7/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Joe Duarte #2	T3249019B	Electrical Conductivity (EC)	EPA 120.1 and	0.729	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	21.7	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	467	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.6	95.8 (T3249019A)	93.4 (T3249019A)	2.53
Electrical Conductivity (EC)	94.4			0.100
Nitrate Nitrogen (NO3-N)	100	95.6 (W3248007A)	92.9 (W3248007A)	2.88

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Grower

Consultant:
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P.O. Box 582085
Modesto, CA 95358

PURCHASE ORDER:
RECEIVED DATE: 9/6/23 12:20 PM
SUBMITTED BY: Irene
ANALYZED DATE: 9/7/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 11.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Deep Well	T3249019E	Electrical Conductivity (EC)	EPA 120.1 and	1.19	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	42.6	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	761	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.6	95.8 (T3249019A)	93.4 (T3249019A)	2.53
Electrical Conductivity (EC)	94.4			0.100
Nitrate Nitrogen (NO3-N)	100	95.6 (W3248007A)	92.9 (W3248007A)	2.88

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